

24ITT32 - Programming in Java

120 Java Lab Tasks and Mini Projects

UNIT I: Classes and Objects (Tasks 1-25)

Basic Java programs, arrays, matrices, classes, objects, methods, student and bank account systems.

UNIT II: Constructors and Nested Classes (Tasks 26-45)

Constructors, this keyword, recursion, static, nested classes, command-line arguments, inventory system.

UNIT III: Inheritance, Packages and Interfaces (Tasks 46-65)

Inheritance, overriding, packages, interfaces, payroll, vehicle and hospital management systems.

UNIT IV: Exception Handling, Multithreading and I/O (Tasks 66-85)

Exception handling, user-defined exceptions, threads, synchronization, file handling and student record systems.

UNIT V: String Handling, Generics and Collections (Tasks 86-110)

Strings, StringBuffer, Generics, ArrayList, LinkedList, HashMap, Stack, Queue, Collections framework.

Mini Projects (111-120)

Student Record Management, Library Management, Banking System, Hospital Management, Quiz App, Inventory Management, Employee Payroll, Hotel Reservation, Contact Management and Course Registration System.